

Amr Essam

AI Engineer

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SUMMARY

Driven and Responsible Junior **Computer Engineer, ITI Graduate**, specialized in Data Science, AI Agents and Python Full Stack Development, Skilled in developing end-to-end systems that integrate machine learning models and Generative AI tools with scalable backend and interactive front-end applications.

WORK EXPERIENCE

[Military Service] Armed Forces Engineering Authority – DB & Oracle APEX Developer (Dec 2024 – Jul 2025)

- Designed and managed Oracle database systems.
- Developed and deployed modern web applications using Oracle APEX and dashboards. [Demo](#)
- Migrated and maintained critical IT Linux systems (domains, servers, applications).

Ericsson – AI Trainee (Jun 2023 – Jul 2023)

- Analyzed telecom datasets (+2 million records) to improve 4G performance.
- Applied ML techniques for network performance and market insights.
- Created visualization dashboards and presented findings to stakeholders using Power BI.

NTI – AI Trainee (Jul 2023 – Aug 2023)

- Actively participated in workshops and group projects.

EDUCATION

- **ITI – ICC (4-month) Full Stack Web Development Python** (Jul 2025 – Nov 2025)
- **Bachelor of Computer and Systems Engineering, Zagazig University** (2019 – 2024)

PROJECTS

- **Retail Analytic Copilot (December 2025):** Full Stack Synthesized Hybrid AI Agent combining RAG & Natural Language to SQL Query to answer retail analytics questions using LangGraph & DSPy with Gemini Model, deployed with Fast API and React JS. With MLflow to trace nodes input/output. [Backend](#) [Frontend](#) [LiveDemo](#)
- **Bash DBMS (September 2025):** Database Management System using Bash Scripting. [Link](#)
- **Soci-Eye (May 2024):** NLP model integrated with mobile application and backend, helps individuals/companies track customer feedback on their new products/campaigns across the internet. [Link](#) [Demo](#)
- **Vision Guard (July 2024):** AI-powered autonomous monitoring robot using image processing techniques [link](#)
- **AI Tracker Gun (April 2023):** Image-processing-based targeting system with Arduino. [link](#)
- **IOT LED (December 2022):** ESP32-based remote LED control with a mobile app. [Demo](#)
- **NASA Space Apps Hackathon (October 2023):** Collaborated on a solution for forest fire detection.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, Bash scripting.
- **Machine learning** algorithms: Supervised, Unsupervised Learning.
- **NLP:** sentiment analysis, Tokenization, Transformers, Hugging face, Vector database.
- **Deep Learning** (PyTorch, Keras).
- Statistics, Probability, Pandas & NumPy.
- MLOps (MLflow, version control (Git)).
- **Database** SQL (Oracle, MySQL, Postgres).
- OOP, Algorithms & Data structure.
- **GenAI:** RAG techniques, Langchain, LangGraph.
- **Web Development:**
 - **Frontend:** HTML5, CSS, Bootstrap, React.
 - **Backend:** Flask, Django, REST API, Fast API
- **OS & Networking:** Linux/RedHat system One, networking fundamentals, Operating system.
- **DevOps technologies:** Containers (**Docker**), Exposure to (SDLC, Scrum, **Agile**, **Jenkins**, Apache)

LANGUAGES

- Arabic (Native)
- English (Advanced)

CERTIFICATES

- Supervised Learning with scikit-learn , [DataCamp](#)
- Unsupervised Learning in Python , [DataCamp](#)
- IBM - Machine Learning with Python, [Coursera](#)
- IBM - Introduction to Deep Learning & Neural Networks with Keras, [Coursera](#)

SOFT SKILLS

- Self-motivated.
- Planning & organizing.
- Attention to details.
- Critical & analytical thinking.
- Communication & Teamwork abilities
- Adaptable.
- Problem solver